EE4-27
EE9-CS5-2
EE9-FPN2-08

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected Copy

SYSTEMS IDENTIFICATION

Correction at 11.45 am Q3 (see first page)

Thursday, 16 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : F. Teng

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1. Consider the stochastic system described by the block-scheme shown in Fig. 1.1 where a_1, a_2, and a_3 are constant parameters. Moreover, $v_1(\cdot) \sim WN(0,1)$ is a white stochastic process and $v_2(\cdot) \sim WN(1,1)$ is a non-zero mean white stochastic process (that is, it is a white stochastic process to which a constant has been added) assumed un-correlated with $v_1(\cdot)$.

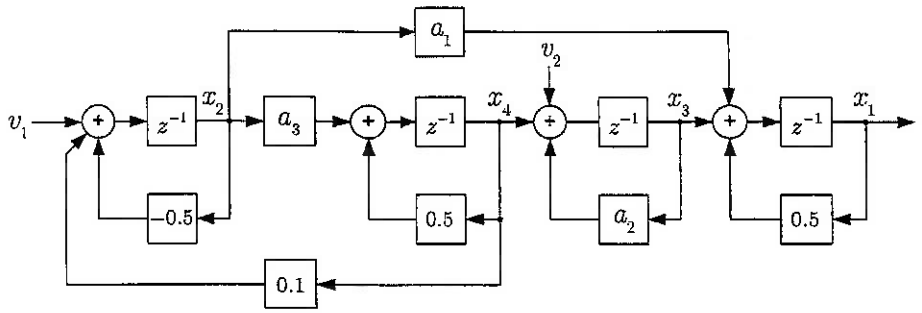


Figure 1.1 Block-scheme of the stochastic system of Question 1.

a) By defining

$$x(t) = \begin{bmatrix} x_1(t) & x_2(t) & x_3(t) & x_4(t) \end{bmatrix}^\top, v(t) = \begin{bmatrix} v_1(t) & v_2(t) \end{bmatrix}^\top,$$

determine suitable matrices A and B such that the stochastic system in Figure 1.1 is represented by the discrete-time state equation:

$$x(t+1) = Ax(t) + Bv(t).$$

[3 Marks]

b) Consider the stochastic processes $x_1(\cdot)$, $x_2(\cdot)$, $x_3(\cdot)$, and $x_4(\cdot)$ generated by the stochastic system in Figure 1.1. Establish conditions on the constants a_1, a_2, and a_3 such that the stochastic processes $x_1(\cdot)$, $x_2(\cdot)$, $x_3(\cdot)$, and $x_4(\cdot)$ are stationary.

[6 Marks]

c) Set $a_1 = a_2 = a_3 = 0$. Compute the expected values $m_1 = \mathbb{E}[x_1(t)]$, $m_2 = \mathbb{E}[x_2(t)]$, $m_3 = \mathbb{E}[x_3(t)]$, and $m_4 = \mathbb{E}[x_4(t)]$ of the stochastic processes $x_1(\cdot)$, $x_2(\cdot)$, $x_3(\cdot)$, and $x_4(\cdot)$, respectively.

[3 Marks]

d) Set $a_1 = a_2 = a_3 = 0$. Using the expected values computed in your answer to Question 1-c), compute the entries $\lambda_{22} = \mathbb{E}[(x_2 - m_2)^2]$, $\lambda_{23} = \mathbb{E}[(x_2 - m_2)(x_3 - m_3)]$, and $\lambda_{24} = \mathbb{E}[(x_2 - m_2)(x_4 - m_4)]$ of the covariance matrix of the stochastic process $x(\cdot) = \begin{bmatrix} x_1(\cdot) & x_2(\cdot) & x_3(\cdot) & x_4(\cdot) \end{bmatrix}^\top$.

[8 Marks]

2. Consider the stochastic process $y(\cdot)$ generated by the state equations below:

$$\begin{aligned} x_1(t) &= 0.5x_1(t-1) + x_2(t-1) + v(t), \\ x_2(t) &= -0.1x_2(t-1) + 0.1x_1(t-1), \\ y(t) &= x_1(t) \end{aligned}$$

where $v(\cdot) \sim WN(0,1)$ is a white process.

a) Determine a suitable ARMA model describing the stochastic process $y(\cdot)$ (*Hint*: combine the above two state-space equations to get an input-output model representation for $y(\cdot)$).

[3 Marks]

b) Compute the first three samples of the correlation function $\gamma_y(\tau)$ of the stochastic process $y(\cdot)$.

[5 Marks]

c) Determine the spectrum $\Gamma_y(\omega)$ of the process $y(\cdot)$.

[3 Marks]

d) Write the difference equation representing the optimal one-step ahead predictor $\hat{y}(t+1|t)$ of $y(t)$ from measured values of $y(\cdot)$.

[3 Marks]

e) Write the difference equation representing the optimal two-steps ahead predictor $\hat{y}(t+2|t)$ of $y(t)$ from measured values of $y(\cdot)$.

[3 Marks]

f) For the optimal two-steps ahead predictor obtained in your answer to Question 2-e), calculate the variance of the prediction error $\varepsilon(t) = y(t+2) - \hat{y}(t+2|t)$.

[3 Marks]

3. Consider an ideal inverted pendulum with a mechanical configuration as the one shown in Figure 3.1. A differential model using standard Newton's law takes on the following form:

$$\tau(t) = ml^2\ddot{\theta}(t) + c_1\dot{\theta}(t) + c_2\text{sign}(\theta(t)) + mlg\cos(\theta(t)) \tag{3.1}$$

where $\theta(t)$ denotes the angular displacement of the pendulum, $\dot{\theta}(t)$ denotes the angular velocity, $\ddot{\theta}(t)$ denotes the angular acceleration, $\tau(t)$ denotes the input torque, g denotes the gravitational acceleration, and m, l, c_1, c_2 are *constant* and *unknown* parameters.

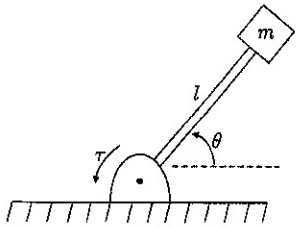


Figure 3.1 An inverted pendulum configuration.

Two ideal sensors are available providing *exact* measurements of $\theta(t)$ and $\tau(t)$, respectively. An experiment to identify the parameters m, l, c_1, c_2 is set up and M pairs of measurements $\{\theta_m(j), \tau_m(j),\ j = 1, 2, \ldots, M\}$ of $\theta(t)$ and $\tau(t)$ at suitable time-instants are collected. Moreover, the following approximate evaluations of the unmeasurable quantities $\dot{\theta}(t)$ and $\ddot{\theta}(t)$ are used:

$$\dot{\theta}(t) = \theta(t+1) - \theta(t),$$
$$\ddot{\theta}(t) = \theta(t+2) - 2\theta(t+1) + \theta(t).$$

a) Provide a discrete-time description of the differential model (3.1).

[4 Marks]

b) By considering the discrete-time model obtained in your answer to Question 3-a), rewrite it in the form of a parametric model characterised as follows:

$$\tau(t) = \sum_{i=1}^{N} \alpha_i p_i(t)$$

for a suitable choice of N, of computable functions $p_i(t), i = 1, 2, \ldots, N$ and where the vector of unknown parameters $\alpha = [\alpha_1, \alpha_2, \ldots, \alpha_N]^\top$ has to be estimated based on the available information and measurements.

[8 Marks]

c) Using the parametric model obtained in your answer to Question 3-b), devise a least-squares procedure to estimate the unknown parameters m, l, c_1, c_2.

[8 Marks]

4. Consider the following set of samples of a stochastic process $v(\cdot)$:

$$v(0) = 10, v(1) = 20, v(2) = -5, v(3) = 3.5, v(4) = -2.5, v(5) = 1.7.$$

a) Only using the samples $v(t), t = 0, 1, \ldots, 5$, show that a MA model of the form $v(t) = a\xi(t) + \xi(t-1), \xi(t) \sim WN(0,1)$, cannot describe the process $v(\cdot)$ (*Hint*: compute the empirical estimates of the correlation function).

[4 Marks]

b) Consider the following AR(1) model:

$$v(t) = av(t-1) + \xi(t), \xi(\cdot) \sim WN(0,1).$$

Based on a numerical procedure and using the given samples $v(t), t = 0, 1, \ldots, 5$, determine an empirical estimation of the unknown parameter a.

[6 Marks]

c) Consider the following AR(2) model:

$$v(t) = av(t-1) + bv(t-2) + \xi(t), \xi(\cdot) \sim WN(0,1).$$

Based on a numerical procedure analogous to the one of your answer to Question 4-b) and using the given samples $v(t), t = 0, 1, \ldots, 5$, determine an empirical estimation of the unknown parameters a and b.

[6 Marks]

d) Using the samples $v(t), t = 0, 1, \ldots, 5$, compute a performance index for the models obtained in your answers to Question 4-b) and 4-c) and compare the two computed performance indexes. Comment on your findings.

[4 Marks]

Syst. Id.

Meaning of: $sign(\theta(t))$:

$sign(\theta(t)) = +1$, if $\theta(t) > 0$

$sign(\theta(t)) = 0$, if $\theta(t) = 0$

$sign(\theta(t)) = -1$, if $\theta(t) < 0$